1 Write queries

Write a query someone could ask, from documents in the corpus

Task description: **SciFact** verifies scientific claims with paper abstracts

3 SAMPLED DOCUMENTS

- Effect of losartan on...
- Alcohol Sales and Risk of...
- Chronic pancreatitis is...



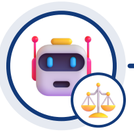
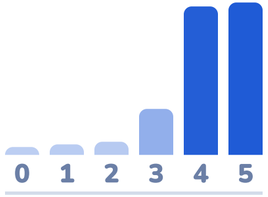
→ 🔎 **What non-genetic factors can trigger cancer development in the presence of oncogenic mutations?**

kept if 3 to 4,000 characters; 100 queries per task

2 Judge the results

Each model retrieves its top 10, and an LLM judge grades them

POINTWISE (DEFAULT)

query + Chronic pancre... →  → 

$\text{grade} = \Sigma\, g \cdot p(g) =$ **4.3** its probability for each grade 0–5

PAIRWISE



shown in both orders, A first and B first, and averaged

3 Rank the models

Turn the judgments into one score per model

POINTWISE

$$\text{nDCG@10} = \frac{\Sigma_{i \le 10}\, \text{grade}_i \,/\, \log_2(i+1)}{\text{same, for the best 10 grades}}$$

mean over queries, then over tasks

PAIRWISE

Bradley–Terry over all wins

one rating per model, on an Elo scale

Gym ranking	MTEB	
Qwen3-Embedding-8B	82.7	#1
Qwen3-Embedding-4B	81.6	#2
Linq-Embed-Mistral	81.2	#3

ρ = 0.95 with MTEB's ranking